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BEHAVIOR TREE-BASED DECISION-MAKING FOR AERIAL THREAT ENGAGEMENT USING A DRONE SWARM

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Abstract. *Autonomous decision-making is a major challenge to the design of multi-drone systems that behave in a coordinated manner to solve a complex problem. This paper proposes a behavior tree for autonomous decision-making in a drone swarm, where each agent carries a short-range weapon with limited ammo. The drone swarm is tasked with defending an area against a number of invading kamikaze drones. Each agent in the swarm makes independent decisions and is supported by communications from a central radar station that provides complete situational awareness. Behavior in each drone is determined by an affine utility model with independent variables reflecting the capabilities of the defender relative to the invaders. The utility model coefficients are optimally computed by a genetic algorithm as the ones that maximize the survival time of the swarm. The proposed method is evaluated through Monte Carlo simulations using a physics-based, in-house simulator with a photorealistic interface developed in Unity 3D. The results show its effectiveness.*

Keywords: *decision-making, multi-drone system, behavior tree, cooperative threat engagement.*

1 INTRODUCTION

Technological advances in mechatronics and communications in the last decade have allowed the emergence of multi-drone systems for a diverse range of practical tasks. These systems are often chosen for their fault tolerance, multitasking proficiency, and flexibility (Kamel *et al.*, 2020; Javed *et al.*, 2024), traits particularly suited to cooperative threat engagement. However, large numbers of agents lead to the well-known challenge of achieving coordination (Gao and Li, 2020; Chen *et al.*, 2024). This paper investigates this subject considering its application to a drone swarm that protects an area from flying kamikaze invaders and that, as such, has to interpret real-time information within a geometrically static setting.

A behavior tree is a hierarchical structure to represent the execution of behaviors in an autonomous system. Behaviors are organized into nodes and their codependencies are represented in tree form, with flow going from the tree root to its leaves (Iovino *et al.*, 2022). Its usage as a decision-making structure has its origins in the construction of objects and worlds in computer games (Lim *et al.*, 2010; Champandard and Dunstan, 2012). Because of their scalability and transparency, they allowed artificial intelligence in game programming to be done simultaneously with other segments of the project, eventually rising to industry standard status (Isla, 2005; Merrill, 2013). For similar reasons, they began to appear in engineering systems (Stenius *et al.*, 2022; Yacharern *et al.*, 2024).

Decision-making in micro air vehicles (MAVs) often takes the form of planning, *i.e.* producing optimal strategies in advance for resource and/or task allocation. Such planning algorithms are available in various formulations, including optimal control (Walton *et al.*, 2022), potential functions (Jing and Wang, 2019), non-convex (Sun *et al.*, 2019) or multi-criteria optimization (Georgiadis *et al.*, 2024), invariably under no or simple dynamic modeling, mostly to avoid prohibitive computational costs. In cooperative threat engagement, however, real-time decisions are usually a requirement due to constantly changing environments. To tackle this problem, Wan *et al.* (2021) and Zhou *et al.* (2022) have proposed separate decision-making strategies based on reinforcement learning in two-dimensional environments under kinematic modeling. Liu *et al.* (2024) have extended those works to a swarm-level learning algorithm in a three-dimensional scenario of direct engagement between opposing drone swarms, with verification in a physics-based simulator implemented in Unity 3D. Avoiding the inherent unpredictability of machine learning designs, Ricardo Jr *et al.* (2023) independently developed a physics-based simulator to investigate the problem of protecting a leader drone in a designated area from

aerial kamikaze threats, with decisions made by a behavior tree with a straightforward, distance-based allocation model under a leader-follower, formation-flight architecture.

Based on the aforementioned gaps, this paper investigates a behavior tree solution to achieve coordination of a swarm of MAVs tasked with defending a designated area from kamikaze aerial threats. We extend the work by Ricardo Jr *et al.* (2023) by presenting a behavior tree design with utility considerations and a leaderless implementation. Each swarm agent does its own real-time decision-making in a dynamic environment with information support from a central radar station. Behavior in each drone is determined by an affine utility model. Design choices and preferences are clearly communicated in the tree's organization and white-box utility modeling. Future-proofing is ensured by a modular structure that allows for simple behavior activation and suppression in different scenarios and development stages.

The following sections are organized as outlined: Section 2 delimits our problem, Section 3 contains this work's methods, Section 4 features simulation parameters and results, and Section 5 offers our conclusions.

2 PROBLEM DEFINITION

This paper deals with decision-making in drone swarms for protecting a critical area. We assume that the defending swarm is homogeneous and leaderless, *i.e.* all drones have exactly the same capabilities. Furthermore, all incoming threats are also drones whose goal is to infiltrate the protected area and do a kamikaze attack against the area itself.

Kamikaze drones are strictly unarmed but are still capable of taking out defenders by direct collision. They move in beelines towards the center of the protected area but are otherwise equipped with the same lower-level guidance and control systems as the defenders and subject to the same dynamic modeling. Whenever one is neutralized, another appears at random in a ring larger, yet comparable in size, than the detection zone. Sucessful infiltration by any kamikaze in any time interval is catastrophic.

Defending drones are permitted to fire at enemies within a no-fly-zone and are each equipped with a simple, short-range projectile weapon. Each shot from this weapon has a chance to miss its target, which we assume to be independent of the distance between target and shooter. We assume there is a central radar station capable of providing situational awareness, so that, once a kamikaze comes within a detection range of the protected zone, its real-time position is broadcast to the swarm. We also assume each defending MAV is capable of visually assessing the capabilities of any kamikaze it approaches and of transmitting this information to the rest of the swarm through the central station.

Each swarm agent is a standalone entity, carrying its own behavior tree and making decisions with only the information set available to it. Communication between agents and the central station is assumed to exhibit low latency and no packet loss. The center of the protected zone is the origin of the position reference frame for all agents and does not drift. It also coincides with the center of both detection and no-fly-zones, as shown in Figure 1.

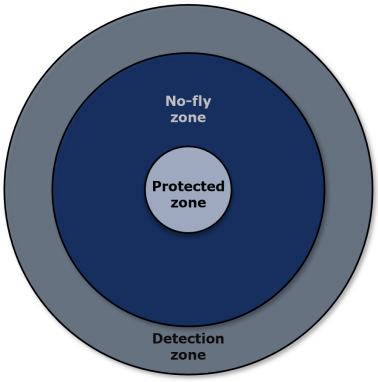


Figure 1. Problem geometry. Infiltration into the inner protected zone is to be avoided. Defenders may only fire at drones inside the no-fly zone. Enemy drones within the detection zone have their positions broadcast to the swarm. Detection zone is taken as larger than the no-fly zone without loss of generality.

3 METHODS

This section introduces our methodology. Section 3.1 discusses dynamic modeling in the simulated environment. Section 3.2 concerns our lower-level guidance and control systems. Section 3.3 and section 3.4, respectively, discuss our situational awareness breadth and the three-dimensional (3D) graphics engine. Section 3.5 goes into the decision-making itself and presents our behavior tree.

3.1 Drone and environmental physics

Both friendly and kamikaze drones are physics objects subject to both translational and rotational dynamics. Earth is modeled as a sphere and all vehicles experience uniform gravity in the radial direction. Drones are subject to three-dimensional disturbances exclusively on torque.

All drones are quadcopters with four identical, fixed rotors. Rotor dynamics are modeled by a first-order differential model with a time constant comparable to simulation step time.

3.2 Low-level control

Lower-level control is replicated from previous papers associated with the CTEDS project - for instance, see Ricardo Jr *et al.* (2023). Those are simulated separately for each MAV and guidance, navigation and control (GNC) systems are divided into two levels. The decision-making algorithm functions outside this structure and inputs a position setpoint. Figure 2 summarizes this architecture.

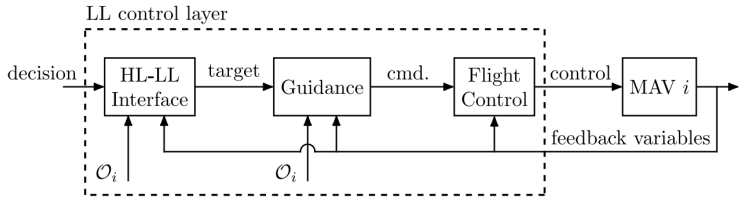


Figure 2. GNC architecture for a single multicopter. $\mathcal{O}_i$ is the set of known information about other agents' states. From Ricardo Jr *et al.* (2023).

Guidance uses a reciprocal acceleration-velocity obstacle (AVO) method to avoid drone and environmental collisions. Collisions with enemies are further avoided by positioning choices from the outer loop input. Flight control is done by a joint attitude-position first-order global sliding mode controller (GSMC). The controller also generates rotor-level throttle inputs that, as a group, closely mirror the original thrust command. Low-level control is the same for both the swarm and the invading kamikaze drones, except that kamikazes exclusively avoid collisions with the environment.

3.3 Situational awareness

Situational awareness data is sent and received in discrete intervals and the entire swarm decides based on the same data packet. Each data packet consists of (a) position estimates for each detected enemy, (b) threat estimates for each visually inspected enemy and (c) information about the most recent decision taken by each agent in the swarm. Packets are sent and received immediately prior to each round of decision-making and are never lost. As we go through the swarm in each round, we visit each agent in a set order that does not depend on its current state. This very order is used as a tiebreaking procedure were two agents to assign themselves to the same task. If this were to happen, the earliest member is this order is given the task and the rest are unassigned.

3.4 3D visualization

Visualization is done through an in-house MATLAB application (Ricardo Jr *et al.*, 2023) capable of simulating physics and GNC for multiple drones in a chosen environment, coupled with a visualization engine in Unity 3D. This code can be further coupled with an external decision-maker, leading to the architecture of Figure 3. This paper improves on Ricardo Jr *et al.* (2023) by integrating all control systems into MATLAB, thereby simplifying the high-low level interface and removing difficult to setup and computationally costly ROS 2 middleware. The simulator runs in real-time and relays MATLAB-based position and velocity information to an exclusively graphical engine in Unity 3D.

3.5 Decision-making

3.5.1 Scenario assessment

The external situational awareness assumption leads to a simple strategy when no threats are present. In this case, it is trivially optimal for the drones to remain stationary when no action is required, rather than patrol the area. It is also useful for the swarm to remain geometrically balanced, that is, distributed in the same shape as the expected probability distribution of incoming threats, for that leads to lowest possible average response time. In our uniformly-distributed case, this leads to an idle swarm positioned at the vertices of regular polygons. Polygon size is a system parameter to be picked by the designer.

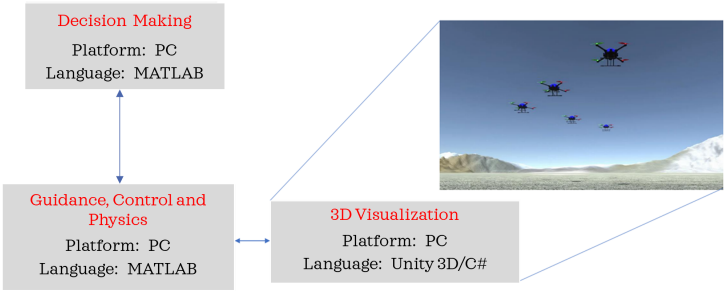


Figure 3. Control systems architecture.

After an attacker is detected, sighted by a swarm member and its capabilities have been assessed, a threat is assigned to it. It is an estimate, given in seconds, of the danger the invader, individually, poses to the defensive system and is defined by Eq. (1).

$$T = t_{eng} - t_{exp} + C \tag{1}$$

where the first term is its engagement time, that is, the time required, under continuous fire, for a defender to neutralize this threat. The second is the exposure time, which is the ratio of double the range of a defender's weapon and the attacker's top speed. This is a conservative estimate of how long that attacker will be exposed to fire, for it assumes the defender cannot follow it. Lastly, there is a constant term.

The result of Eq. (1) must be further capped below at zero, for negative threat terms are unacceptable, as they would imply that an attacker's sighting lowers the danger to the defence. The constant is, then, a design parameter to define the weakest enemy capable of altering system behavior. In this text, we shall use $C = 10$ s. The sum of the threats of all sighted enemies is a measure of overall stress on the defence and will be called the composite threat.

3.5.2 Behavior

We propose a behavior tree in three branches, each representing a specific situation (or mode) in the course of operations. The tree's outline and behavioral hierarchy is shown in Fig. 4. An external comms module exists outside of this structure and overlays decision information to other swarm members.

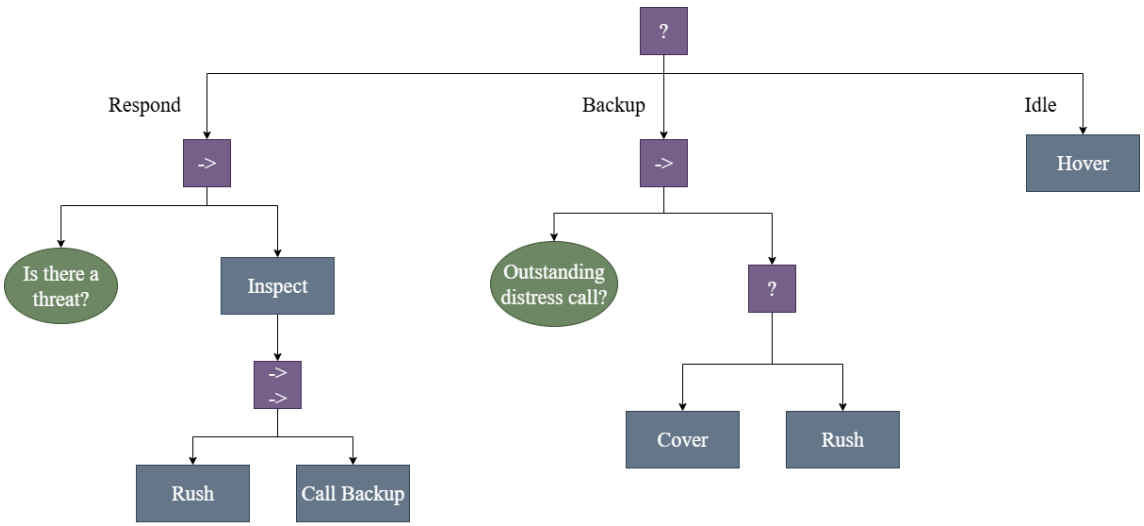


Figure 4. Behavior tree outline.

The idle branch governs the polygon-making behavior of Section 3.5.1. Each drones' position set point is the unit vector diametrically opposed to the swarm's instantaneous center of gravity multiplied by the chosen radius.

The backup branch responds to distress calls when a particularly strong enemy is sighted by another defender. It is the receiver part of the *call backup* behavior of the respond branch. It allows for a full-on response (*rush*) or for cover behavior, where the backup provider leaves the idle stance and moves in the direction of the caller, but without fully committing. This branch allows the swarm to deal with particularly resistant enemies.

The respond branch contains a intermediate node, *inspect*, where the MAV moves out from its idle position to perform visual assessments once an enemy is detected. It can, then, opt to *rush* aggressively towards its target, to send a distress call for backup or to just wait. A distress call is sent whenever it calculates an enemy engagement time greater than its expected exposure. The call is outstanding until the exposure time exceeds the engagement time of the responding mass of drones.

Rushing is inherently risky, for it introduces geometric vulnerabilities by leaving unguarded the space previously occupied by a defender. On the other hand, it lowers enemy survival time and, consequently, the risk that the swarm is numerically overwhelmed. We compute its utility (Eq. (2)) by an affine model in three variables (heuristics), namely, defender travel time, threat of the enemy to be rushed, and composite threat, that is, the total threat of sighted enemies. The constant term of the affine model is a risk aversion parameter. The defender decides to rush whenever the computation of Eq. (2) is greater than zero.

$$U_{rush} = c_1 \frac{R_{target} - R_{inspect}}{v_{max,friendly}} + c_2 T_{target} + c_3 T_{composite} + c_4 \tag{2}$$

where U_{rush} is the rush utility, R_{target} and $R_{inspect}$ are, respectively, the target's distance from the protected area center and the distance at which visual assessments are performed within *inspect* behavior (set as a system parameter), $v_{max,friendly}$ is the deciding defender's maximum speed, T_{target} is that target's threat, $T_{composite}$ is the composite threat and $\{c_n\}$ is a set of model coefficients.

The set of c_n values is a system parameter that depends on problem geometry and enemy characteristics and distribution. In the next section, we shall propose a solution by genetic algorithm selecting for swarm survival time.

4 SCENARIO SIMULATION AND RESULTS

This section presents the simulation setup and results. Section 4.1 lists problem geometry and defending drone specifications. Section 4.2 introduces test case parameters. Section 4.3 provides a summary and a discussion of the corresponding results.

4.1 Parameters

Simulation parameters and dimensions are presented in the following tables. Table 1 contains dimensions for the problem geometry (see Fig. 1). Table 2 contains a summary of defender specifications. Values in both tables are normalized by defender maximum speed whenever applicable. Thus, a radius of one second means a defender takes one second to move across that distance at maximum speed.

Table 3 lists behavior-specific values. Inspect and idle radii are the set point distances for each behavior in relation to the protected zone center and are chosen at once. The remaining values are the coefficients of the rush utility function (Eq. (2)). These values are picked by a genetic algorithm selecting for swarm survival time in a scenario with a swarm of $N_1 = 3$ agents against $N_2 = 6$ enemies capable of withstanding four shots each and of moving at a maximum speed 50% greater than that of the defenders.

Thus, the generic algorithm considers the problem of

$$\max_{x \in \mathbb{R}^5} \; t_{survival} \quad \text{subject to} \quad x_5 > 0 \tag{3}$$

where $x = \{c_1, c_2, c_3, c_4, R_{inspect}\}$, that is, the four affine model coefficients and the inspect/idle radius, in this order.

Table 1. Scenario dimensions - normalized by defender maximum speed.

Parameter	Value
Protected zone radius	0.5 s
Visual assessment radius	15 s
Detection radius	20 s
No-fly zone radius	20 s

Table 2. Defender characteristics - normalized by defender maximum speed.

Parameter	Value
Radius	0.25 s
Weapon range	4.8 s
Firepower(1)	1.33 s^{-1}
Ammo capacity(2)	100
Hit probability	95%

(1) Shots per second. (2) Per drone.

Table 3. Behavior-specific parameters.

Parameter	Value
c_1	-3.3
c_2	3.6
c_3	1.6
c_4	4.8
Idle radius	2 s
Inspect radius	2 s

The resulting behavior is as follows: defenders gather near the protected zone and rush aggressively whenever an enemy gets close enough. The principal merits of this organization are the ease of uniformization of usage rates between swarm members, even when enemies are concentrated in specific directions, the reduction of directional vulnerability when swarm members move out to close down enemies and the increasing availability of reinforcements as threats move closer to the protected zone. The main drawback is the difficulty of avoiding collisions between agents, especially for large swarms.

4.2 Scenarios

For testing purposes, we will consider the scenario where a swarm of N = 3 MAVs is tasked with defending a point protected zone while outnumbered two-to-one. We propose nine possibilities for enemy strength, varying both fire resistance and top speed. Figure 5 contains all nine test cases.

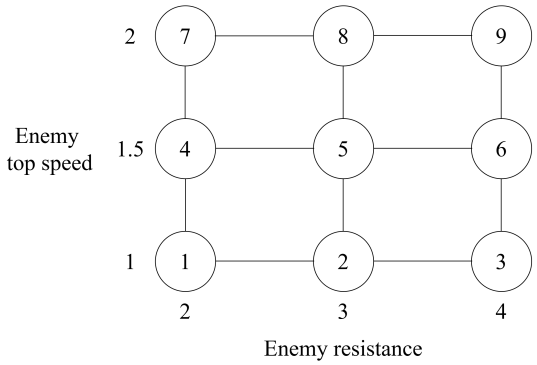


Figure 5. Scenario-specific settings.

Each scenario was simulated $N = 100$ times in a Monte Carlo approach. The variance between simulations is primarily due to the randomness in threat generation. Since the defenders are outnumbered, the case when attackers approach in sync with large angular dispersion is particularly dangerous. A secondary source of randomness is the shot hit probability, set at 95%.

4.3 Results

Figure 6 shows variability in survival time for a sample case. Afterward, Tab. 4 offers a comprehensive summary of simulation results. A defensive effort is deemed successful if at least 90% of available swarm ammunition is expended.

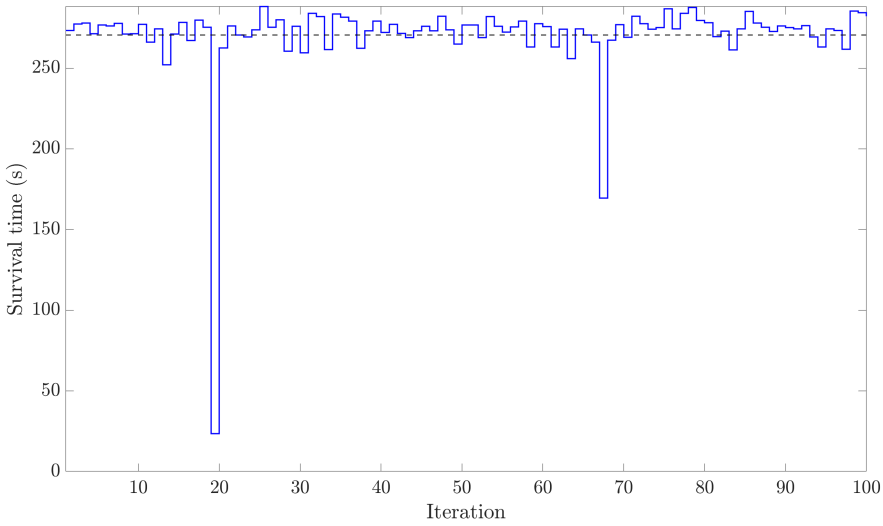


Figure 6. Survival time per iteration in scenario 3.

Table 4. Summary of simulation results. The confidence interval corresponds to a single standard deviation.

	Survival time (s)	Enemies downed	No. of successes	Friendly drones lost
Scenario 1	511.0 ± 71.8	134.5 ± 20.0	86	34
Scenario 2	361.2 ± 16.1	93.9 ± 4.5	99	2
Scenario 3	270.5 ± 27.9	69.4 ± 7.4	98	4
Scenario 4	350.2 ± 14.1	138.6 ± 5.1	96	0
Scenario 5	241.2 ± 16.9	90.9 ± 6.1	92	0
Scenario 6	159.2 ± 53.3	57.1 ± 19.4	66	0
Scenario 7	279.0 ± 31.7	136.9 ± 15.3	95	0
Scenario 8	132.3 ± 65.2	60.8 ± 30.6	37	0
Scenario 9	52.3 ± 36.1	22.0 ± 16.5	5	0

The results of Tab. 4 show that the proposed defensive strategy is effective for the proposed test cases. While the swarm does start getting frequently overwhelmed in later scenarios, this is largely due to the combination of razor-thin margins for error against those setups combined with the variance caused by hit probability.

The main issue with this strategy is illustrated by the collisions in the lower threat scenarios. Looking at iteration-level data, all drone losses came in pairs, which suggests that those collisions involved friendly agents exclusively. This is likely a byproduct of diminished rushing tendencies as weaker attackers lead to lower utilities, all else constant. In those cases, to reduce risk, defenders allow attackers to get closer to the protected zone before moving in. This leads to aggressive maneuvering in tight quarters and stresses the collision avoidance systems. In fact, all unsucessful iterations at the lower speed were due to the crashes, those being particularly common in scenario 1, where enemy threat is lowest.

5 CONCLUSIONS

This paper proposes a behavior tree structure for decision-making in a cooperative threat engagement problem in a drone swarm. In the resulting strategy, drones will usually stay close to the protected zone center and close in decisively on incoming threats. As a verification procedure, nine test cases of increasing aggressiveness were produced in an in-house simulated environment. Testing was done in a Monte Carlo simulation, with 100 iterations per test case, and the results demonstrate the strategy's effectiveness.

The emergent low-threat collision issue is possibly solved by setting larger idle radii, with performance impacts in other scenarios to be determined. Other results show that a BT architecture is capable of providing a flexible defensive strategy that is clearly communicated to the user. Further on, some future steps in this endeavour would be to conduct system appraisals under stressed situational awareness scenarios and against more strategic, cooperative enemy behavior.

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